

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

Yes, Jordan's statement is true. The value a_2 defined by the limit of the given code is $a_2 = 1 + 2 \cdot 72^{-N} - \frac{2 \cdot 72^{-3N}}{1 - 72^{-N}}$ with $N = 10^{24}$, which is strictly greater than 1 but unimaginably close to it. The linear program to minimize total time $x_1 + x_2$ subject to

$x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$ therefore has a unique minimizer at $(x_1^*, x_2^*) = (0, 1/a_2) \approx (0, 1)$.

The infinity-norm distance from $(0.3, 0.7)$ to $(0, 1/a_2)$ is $\max(0.3, |0.7 - 1/a_2|) \approx 0.3$, which is indeed ≤ 0.4 .

The later "revealed" claim about $(0.6, 0.4)$ would give a distance of $\approx 0.6 > 0.45$ and is therefore false under the given a_2 .